

Instrumental Variables

EC 421, Set 11

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Prologue

Schedule

Last Time

Causality

Today

- Review: Causality
- New: Instrumental variables

Causality

Review

Causality

Review

In our last lecture, we returned to the concept of **causality**.

We worked through the *Rubin causal model*, in which we defined

- y_{1i} : the outcome for individual i if she had received treatment
- y_{0i} : the outcome for individual i if she had not received treatment

and we referred to individuals who did not receive treatment as *control*.

If we were able to know both y_{1i} and y_{0i} , we could calculate the causal effect of treatment for individual i , i.e.,

$$\tau_i = y_{1i} - y_{0i}$$

Causality

Review

Fundamental problem of causal inference:

We cannot simultaneously know y_{1i} and y_{0i} .

Either we observe individual i in the treatment group, i.e.,

$$\tau_i = y_{1i} - ?$$

or we observe i in the control group, i.e.,

$$\tau_i = ? - y_{0i}$$

but never both at the same time.

Causality

Review

If we want to know τ_i (or at least $\bar{\tau}$), what can we do?

Causality

Review

Result: We showed that even when the treatment effect is constant (meaning $\tau_i = \tau$ for all i),

$$\begin{aligned} & Avg(y_i \mid D_i = 1) - Avg(y_i \mid D_i = 0) \\ &= \tau + \underbrace{Avg(y_{0,i} \mid D_i = 1) - Avg(y_{0,i} \mid D_i = 0)}_{\text{Selection bias}} \end{aligned}$$

which says that the difference in the groups' means will give us a **biased estimate** for the causal effect of treatment **if we have selection bias**.

Causality

Review

Q: What is this **selection bias**?

Causality

Review

Practical problem:

$$\underbrace{Avg(y_{0,i} \mid D_i = 1)}_{\text{Unobservable}} - Avg(y_{0,i} \mid D_i = 0)$$

(back to the *fundamental problem of causal inference*)

Causality

Review

Example: Imagine we have two people—Al and Bri—and a single binary treatment, college. We are interested in the effect of college on earnings.

$$\text{Earn}_{1,\text{Al}} = \$60\text{K}$$

$$\text{Earn}_{0,\text{Al}} = \$30\text{K}$$

$$\text{Earn}_{1,\text{Bri}} = \$140\text{K}$$

$$\text{Earn}_{0,\text{Bri}} = \$110\text{K}$$

Causality

Review

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The selection bias...

Causality

Review

We have (at least) two problems...

1. Selection bias is difficult to observe
2. If selection bias is present, our estimate for τ is biased, preventing us from understanding the causal effect of treatment.

Instrumental variables

Intro

Recall: **Selection bias** means our **treatment** and **control** groups differ on some unobserved/omitted dimension. (**Endogeneity**)

Instrumental variables

Introductory example

Example We want to estimate the effect of veteran status on earnings,

$$\text{Earnings}_i = \beta_0 + \beta_1 \text{Veteran}_i + u_i \quad (1)$$

We would like to calculate $\text{Earnings}_{1i} - \text{Earnings}_{0i}$, but we can't.

And OLS will likely be biased for (1) due to selection/omitted-variable bias.

Instrumental variables

Introductory example

Imagine that we can split veteran status into an exogenous part and an endogenous part...

$$\begin{aligned}\text{Earnings}_i &= \beta_0 + \beta_1 \text{Veteran}_i + u_i \\ &= \beta_0 + \beta_1 \left(\text{Veteran}_i^{\text{Exog.}} + \text{Veteran}_i^{\text{Endog.}} \right) + u_i \\ &= \beta_0 + \beta_1 \text{Veteran}_i^{\text{Exog.}} + \underbrace{\beta_1 \text{Veteran}_i^{\text{Endog.}}}_{w_i} + u_i \\ &= \beta_0 + \beta_1 \text{Veteran}_i^{\text{Exog.}} + w_i\end{aligned}\tag{1}$$

We could use this exogenous variation in veteran status to consistently estimate β_1 .

Instrumental variables

Introductory example

Q: What would exogenous variation in veteran status mean?

Instrumental variables

Instruments

Q: How do we isolate this *exogenous variation* in our explanatory variable?

Instrumental variables

Instruments

Q: What's an instrument?

A: An **instrument** is a variable that is

1. **correlated** with the **explanatory variable** of interest (**relevant**),
2. **uncorrelated** with the **disturbance** (**exogenous**).

Instrumental variables

Instruments: Relevance

Relevance:

We can actually test this requirement using regression and a t test.

Instrumental variables

Instruments: Exogeneity

Exogeneity:

z_i must be uncorrelated with our disturbance u_i . **Not testable.**

Instrumental variables

Instrumental review

Let's recap...

- Our instrument must be **correlated with our endogenous variable**.
- Our instrument must be **uncorrelated with any other variable that affects the outcome**.

Instrumental variables

Back to our example

For **veteran status** we considered three potential instruments:

If we have a *valid* instrument (e.g., the draft lottery), how do we use it?

Instrumental variables

Estimation

Recall: We want to estimate the effect of veteran status on earnings.

$$\text{Earnings}_i = \beta_0 + \beta_1 \text{Veteran}_i + u_i$$

Instrumental variables

Estimation

Recall: We want to estimate the effect of veteran status on earnings.

$$\text{Earnings}_i = \beta_0 + \beta_1 \text{Veteran}_i + u_i$$

and we know that the draft affected veteran status.

Instrumental variables

Estimation

We just wrote out an expression for the effect of **the draft** on **earnings**, i.e.,

$$\begin{aligned} \text{(Effect of \textbf{the draft} on \textbf{earnings})} = & \\ & \text{(Effect of \textbf{the draft} on \textbf{veteran status})} \times \\ & \text{(Effect of \textbf{veteran status} on \textbf{earnings})} \end{aligned}$$

Instrumental variables

Estimation: Bring it all together

By estimating two regressions involving our **instrument**,

1. The effect of the **instrument** on the **endogenous variable**, e.g.,

$$\text{Veteran}_i = \gamma_0 + \gamma_1 \text{Draft}_i + v_i$$

2. The effect of the **instrument** on the **outcome variable**, e.g.,

$$\text{Earnings}_i = \pi_0 + \pi_1 \text{Draft}_i + w_i$$

Instrumental variables

Estimation: Bring it all together

By estimating two regressions involving our **instrument**,

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$$\text{Veteran}_i = \gamma_0 + \gamma_1 \text{Draft}_i + v_i$$

2. The effect of the **instrument** on the **outcome variable**, e.g.,

$$\text{Earnings}_i = \pi_0 + \pi_1 \text{Draft}_i + w_i$$

we can estimate our desired effect:

$$(\text{Effect of } \text{veteran status} \text{ on } \text{earnings}) = \frac{\pi_1}{\gamma_1}$$

Instrumental variables

Estimation: Bring it all together

So with instrumental variables, we estimate β_1 using

$$\hat{\beta}_1^{\text{IV}} = \frac{\hat{\pi}_1}{\hat{\gamma}_1}$$

where $\hat{\pi}_1$ and $\hat{\gamma}_1$ come from the two equations we just discussed.

Figure 1

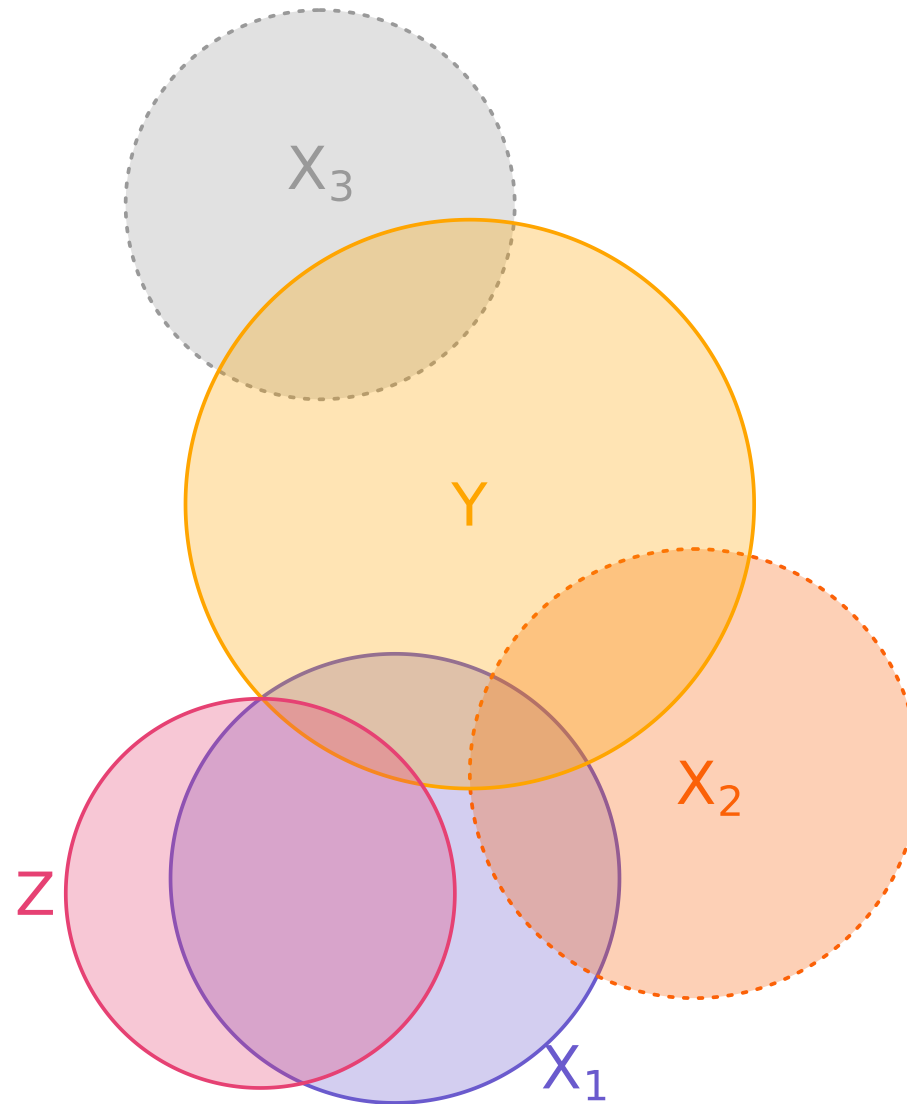


Figure 2

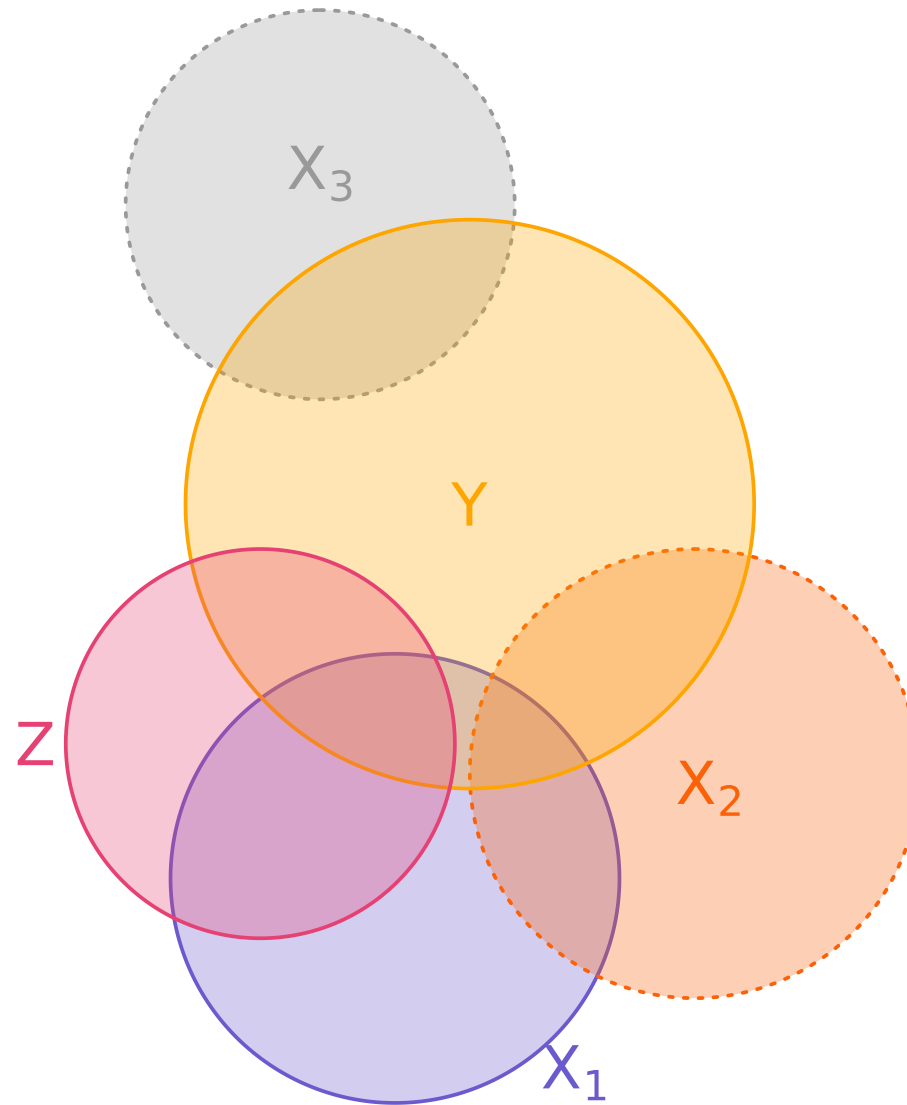


Figure 3

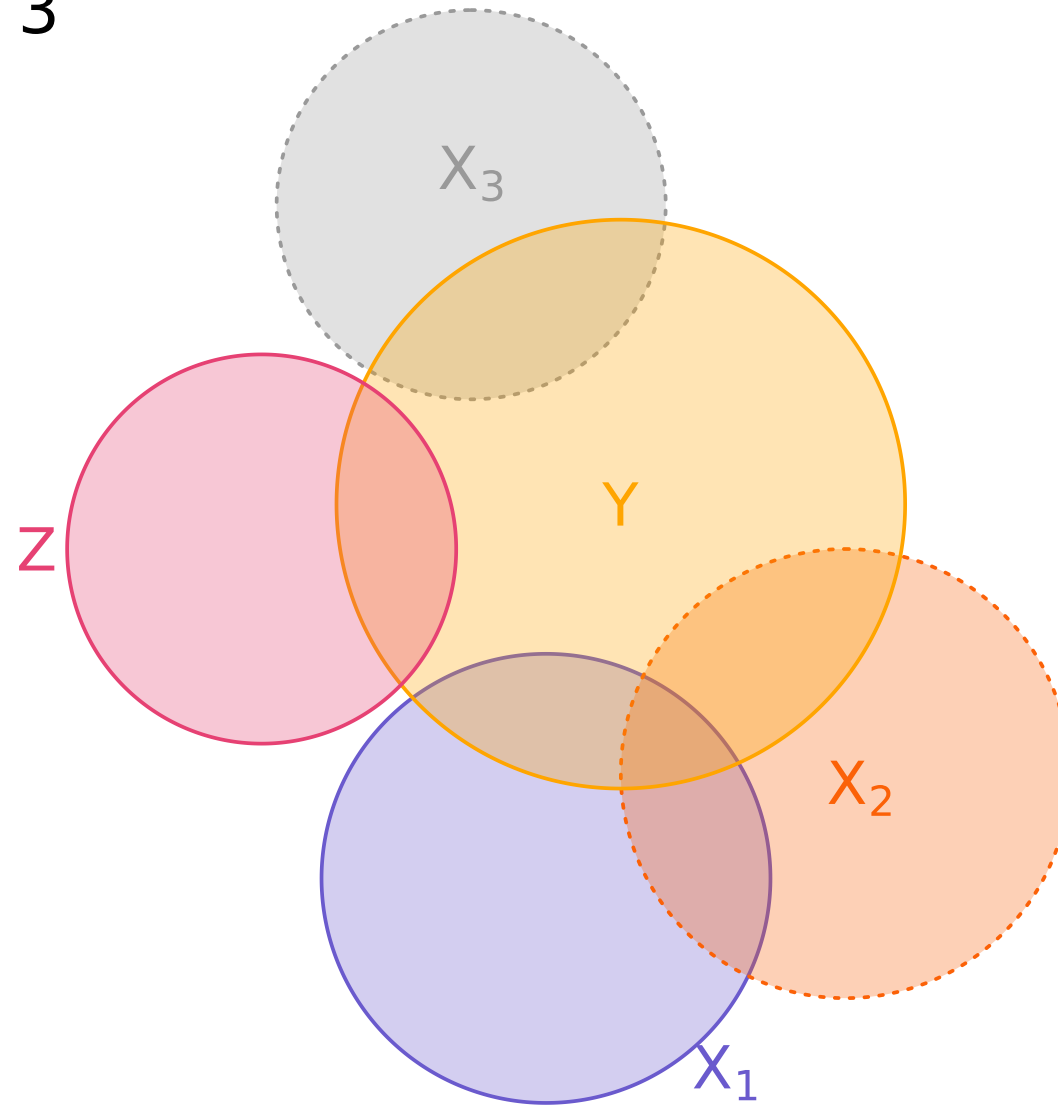
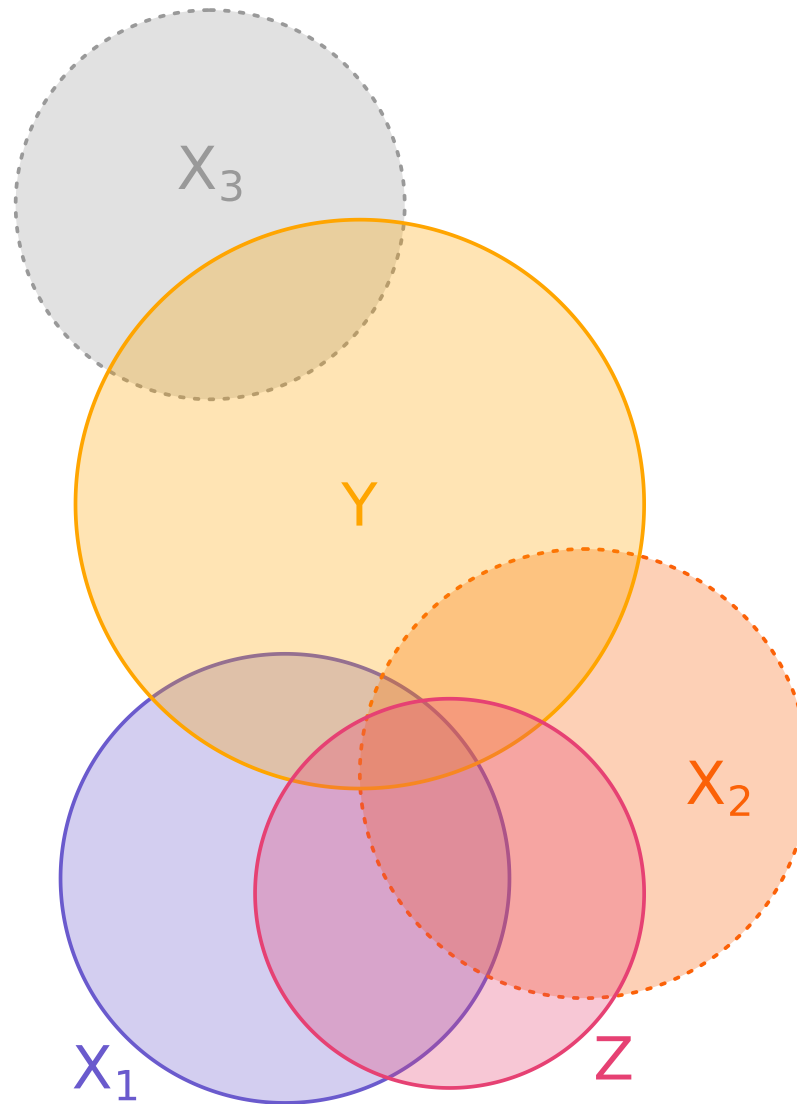


Figure 4



Venn diagram explanation

In these figures (Venn diagrams)

- Each circle illustrates a variable.
- Overlap gives the share of correlation between two variables.
- Dotted borders denote *omitted* variables.

Take-aways

- Figure 1: **Valid instrument** (relevant; exogenous)
- Figure 2: **Invalid instrument** (relevant; not exogenous)
- Figure 3: **Invalid instrument** (not relevant; not exogenous)
- Figure 4: **Invalid instrument** (relevant; not exogenous)

Let's work an example in R.

Instrumental variables

Example in R

Back to our age-old battle to estimate the returns to education.

```
#> # A tibble: 722 × 4
#>   wage education education_dad education_mom
#>   <int>      <int>      <int>      <int>
#> 1   769         12         8         8
#> 2   808         18        14        14
#> 3   825         14        14        14
#> 4   650         12        12        12
#> 5   562         11        11         6
#> 6   600         10         8         8
#> 7  1154         15         5        14
#> 8  1000         12        11        12
#> 9   930         18        14        13
```

Instrumental variables

Example in R

OLS for the returns to education with will likely (definitely) be biased...

$$\text{Wage}_i = \beta_0 + \beta_1 \text{Education}_i + u_i$$

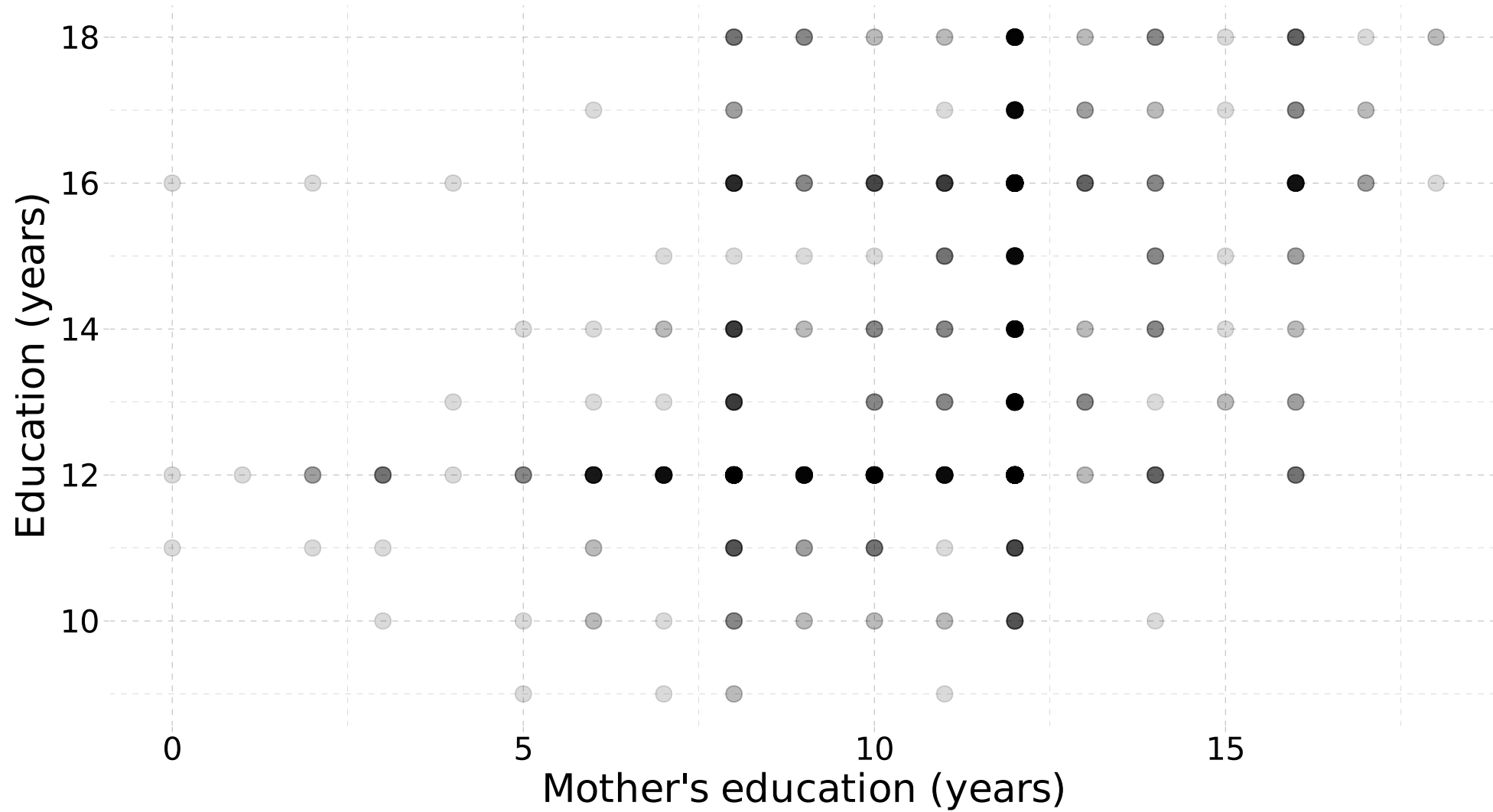
Instrumental variables

Example in R

We can check/test the *relevance* of **mother's education** for **education**.

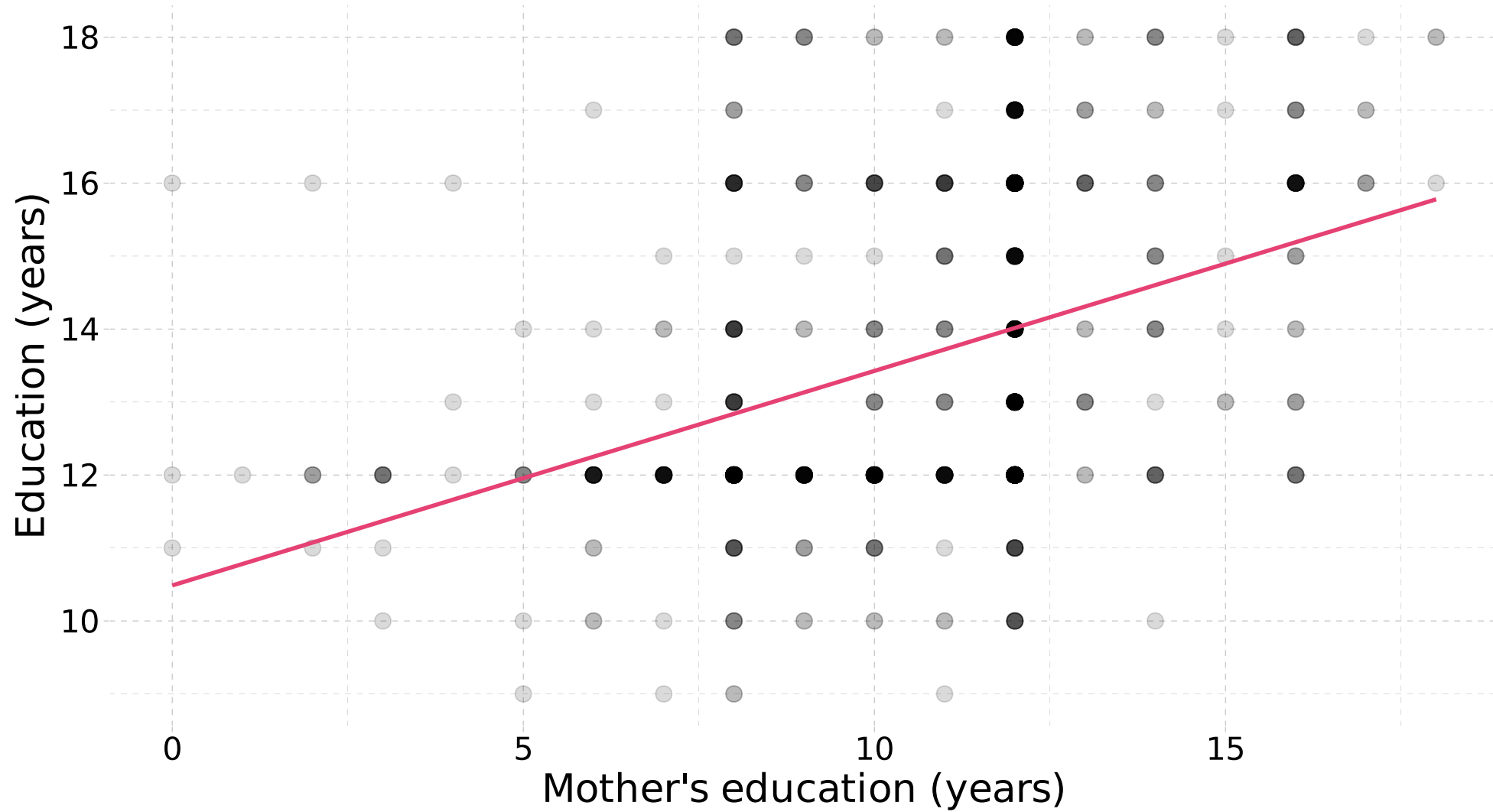
Instrumental variables

Visualizing the first stage



Instrumental variables

Visualizing the first stage



Instrumental variables

Exogeneity

Q: What does **exogeneity** mean in this case?

Instrumental variables

Example in R

Now, let's estimate the **reduced form**:

The effect of the **instrument** on our **outcome variable**.

$$\text{Wage}_i = \pi_0 + \pi_1 (\text{Mother's Education})_i + w_i$$

Instrumental variables

Example in R

So what is our IV-based estimate for the returns to education?

$$\text{Wage}_i = \beta_0 + \beta_1 \text{Education}_i + u_i$$

Instrumental variables

Example in R

Alternative: Use the function `iv_robust()` from the `estimatr` package.

This new function `iv_robust` works very similar to our good friend `lm`:

```
iv_robust(y ~ x | z, data = dataset)
```

- `formula` Specify the regression followed by `|` and your instrument (`z`).
- `data` You still need a dataset.

Note: As you might guess by its name, `iv_robust` calculates heteroskedasticity-robust standard errors by default.

Instrumental variables

Example in R

In practice...

```
# Estimate our IV regression
```

```
iv_est <- iv_robust(wage ~ education | education_mom, data = wage_df)
```

Term	Est.	S.E.	t stat.	p-Value
Intercept	-501.474	226.476	-2.21	0.0271
Education	108.214	16.810	6.44	

Instrumental variables

More

So now we know how to “do” instrumental variables

Instrumental variables

More

So now we know how to “do” instrumental variables *when we have one endogenous variable and one exogenous variable.*

1. Estimate the reduced form (regress **outcome var.** on **instrument**).
2. Estimate the first stage (regress **expl. var.** on **instrument**).
3. Calculate the IV estimate using the estimates from (1) and (2).

Our magical **instrument** isolates the exogenous variation in our **endogenous variable**.

Q: What if we want more? (E.g., more instruments or endog. variables)

A: Too bad. Extend IV to **two-stage least squares (2SLS)**.

Two-stage least squares

Two-stage least squares

Intro

The intuition and insights of IV carry over into two-stage least squares.

Two-stage least squares

Intro

Two-stage least squares is very flexible—we include other controls, additional endogenous variables, *and* have multiple instruments.

But don't get too distracted by this fancy flexibility.

We still need **valid** instruments.

Two-stage least squares

In R

Back to our *returns to education* example.

$$\text{Wage}_i = \beta_0 + \beta_1 \text{Education}_i + u_i$$

Imagine that mother's *and* father's education are both valid instruments.

Then our **first-stage regression** is

$$\text{Education}_i = \gamma_0 + \gamma_1 (\text{Mother's education})_i + \gamma_2 (\text{Father's education})_i + v_i$$

which we can estimate via OLS.

Two-stage least squares

In R

$$\text{Education}_i = \gamma_0 + \gamma_1(\text{Mother's education})_i + \gamma_2(\text{Father's education})_i + v_i$$

```
stage1 <- lm(education ~ education_mom + education_dad, wage_df)
```

First-stage results:

Term	Est.	S.E.	t stat.	p-Value
Intercept	9.845	0.305	32.31	
Mother's Education	0.149	0.032	4.62	
Father's Education	0.216	0.028	7.84	

Two-stage least squares

In R

Using our estimated first stage, we grab the *fitted* endogenous variable

$$\widehat{\text{Education}}_i = \hat{\gamma}_0 + \hat{\gamma}_1(\text{Mother's education})_i + \hat{\gamma}_2(\text{Father's education})_i$$

Two-stage least squares

In R

$$\text{Wage}_i = \delta_0 + \delta_1 \widehat{\text{Education}}_i + \varepsilon_i$$

```
stage2 <- lm(wage ~ education_hat, wage_df)
```

Second-stage results:

Term	Est.	S.E.	t stat.	p-Value
Intercept	-454.683	198.149	-2.29	0.022
Fitted Education	104.789	14.462	7.25	

Ordinary least squares

Term	Est.	S.E.	t stat.	p-Value
Intercept	176.504	89.152	1.98	0.0481
Education	58.594	6.439	9.10	

Instrumental variables

Term	Est.	S.E.	t stat.	p-Value
Intercept	-501.474	226.476	-2.21	0.0271
Education	108.214	16.810	6.44	

Two-stage least squares w/ two instruments

Term	Est.	S.E.	t stat.	p-Value
Intercept	-454.683	198.149	-2.29	0.022
Education	104.789	14.462	7.25	

Two-stage least squares

In R

As you probably guessed, R will do both of the stages for you.

Two-stage least squares

There's more!

Because 2SLS **isolates exogenous variation in an endogenous variable**, we apply it in other settings that are biased from *endogenous* relationships.

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